College Debt Outcome Mismatch Analyzer

Surfacing predatory academic programs using federal data — because students deserve to know what a degree actually costs.

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Data Source	U.S. Department of Education — College Scorecard API
Scope	6,322 institutions analyzed · 4,500 with complete outcome data
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GitHub	github.com/rushil1356/College-debt-analyzer

4,500 **Institutions Analyzed**	**69** **Flagged Predatory**	**3.22** **National Avg Value Score**	**46** **Below Break-even**

1. The Problem

Every year, over **2 million Americans** enroll in college programs without knowing a fundamental truth — some degrees cost more than they will ever pay back. The federal government collects this data. It is public, free, and comprehensive. But it is buried in machine-readable files that no student, parent, or counselor has time to interpret.

This analysis surfaces what that data actually shows — which institutions leave graduates financially worse off, and which deliver genuine value.

The Value Score

This project introduces a single metric — the **Value Score**:

Value Score = Median Earnings (10 yrs out) / Median Debt at Graduation

Range	Meaning	Assessment
Above 1.0	Earn more than borrowed	Healthy
Equals 1.0	Earn exactly what borrowed	Break-even
Below 1.0	Earn less than borrowed	Red flag
National avg	3.22 — most colleges OK	Baseline

2. The National Picture

The headline finding is reassuring: **the American college system is not broken**. The vast majority of institutions provide graduates with reasonable returns. The problem is concentrated, not systemic.

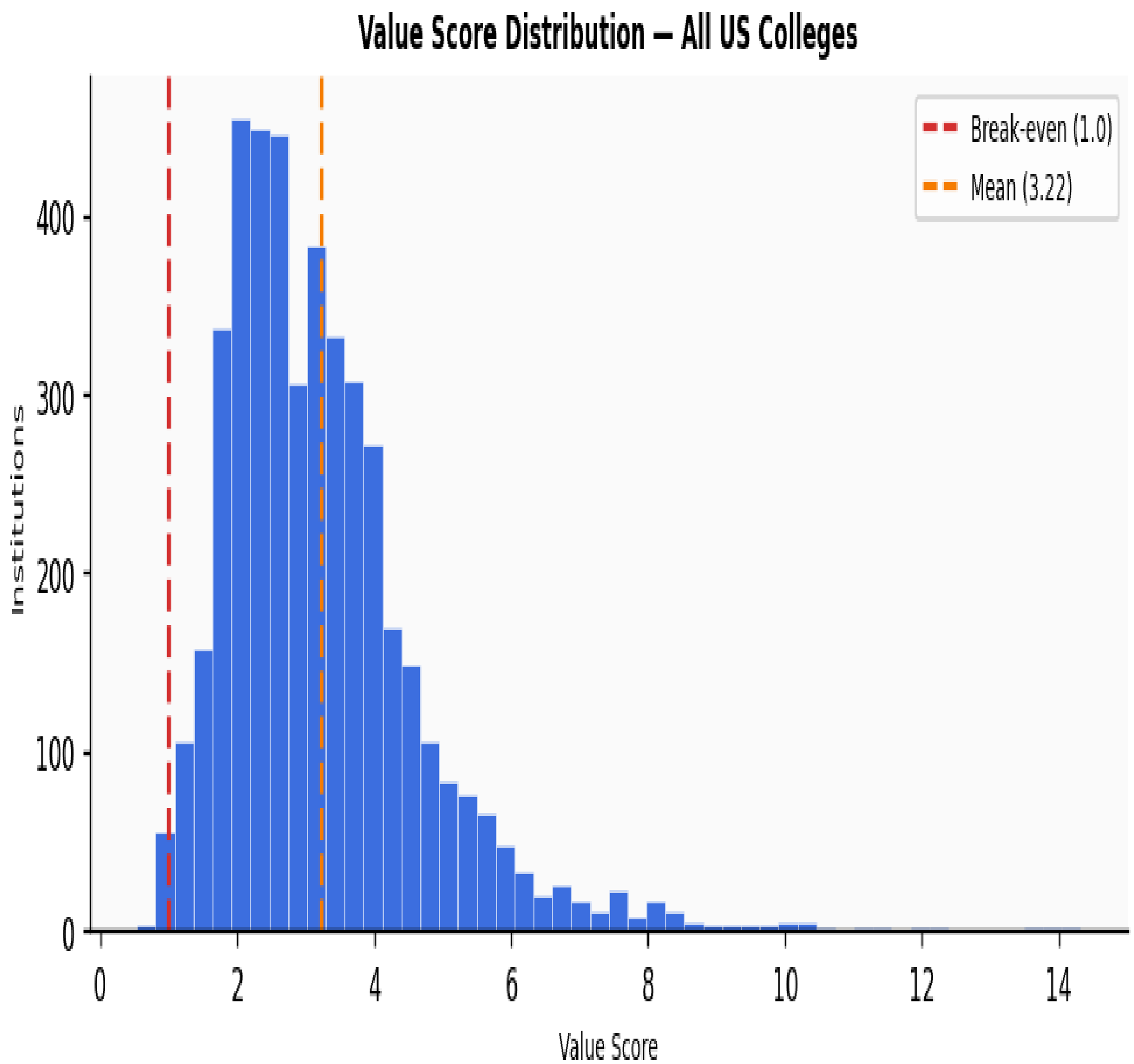


Figure 1 — Right-skewed distribution with mean 3.22. Most schools deliver healthy returns.

- **Mean of 3.22** — average graduate earns $3.22 for every $1.00 borrowed.
- **Right-skewed** — most schools cluster 2–4, with a long tail of high-performing vocational programs.
- **Only 46 institutions (1.0%)** fall below break-even — a concentrated, identifiable problem.

3. Worst Value Institutions

The 20 schools with the lowest value scores represent the clearest cases of financial mismatch — graduates carry debt that equals or exceeds their earnings a full decade after enrollment.

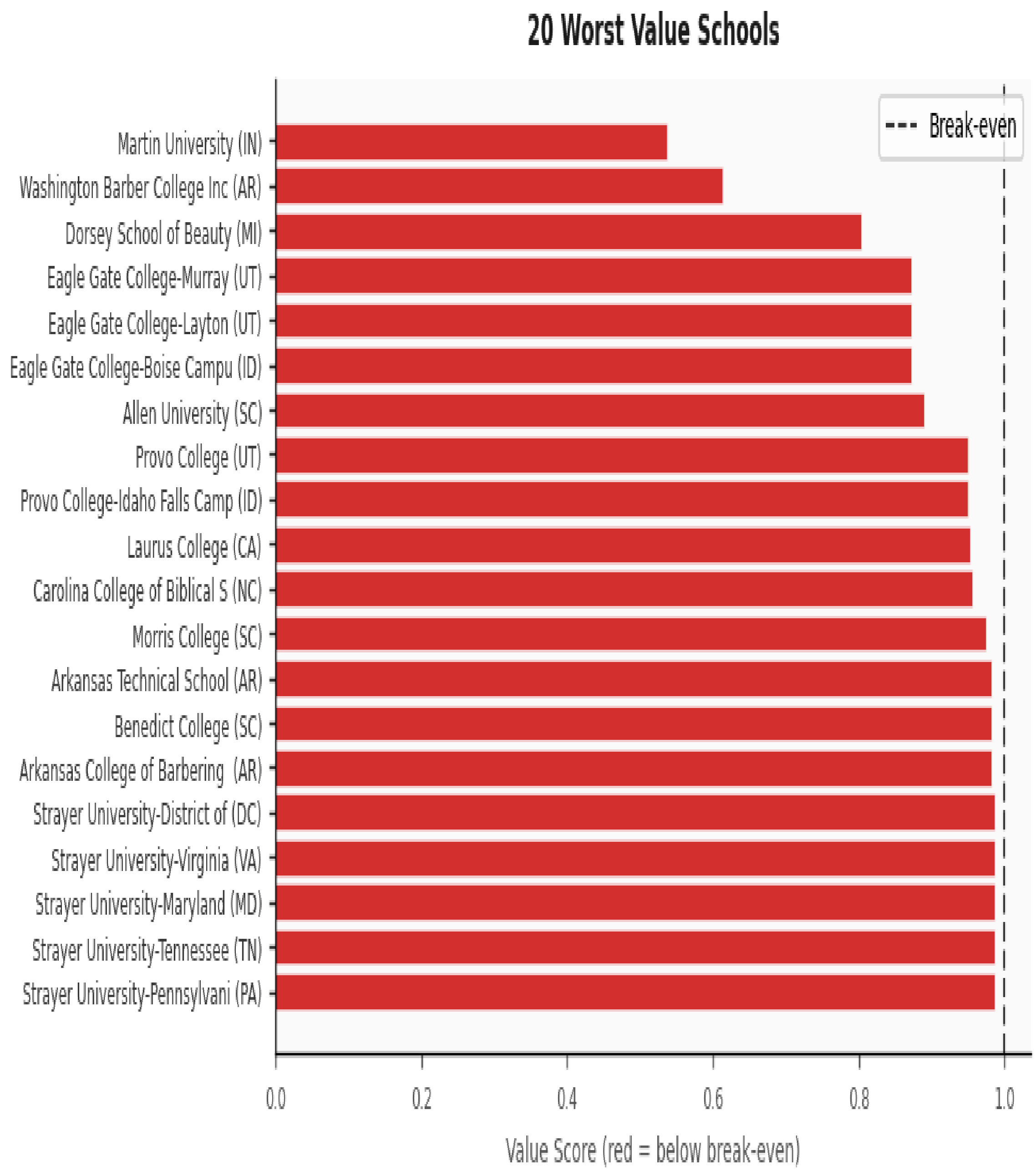


Figure 2 — Red bars indicate graduates earning less than borrowed.

Bottom 10 Schools

School	St	Earnings	Debt	Score
Martin University	IN	$22,544	$42,002	0.537

Washington Barber College Inc	AR	$8,579	$14,000	**0.613**
Dorsey School of Beauty	MI	$13,256	$16,500	**0.803**
Eagle Gate College-Murray	UT	$37,518	$43,021	**0.872**
Eagle Gate College-Layton	UT	$37,518	$43,021	**0.872**
Eagle Gate College-Boise Campus	ID	$37,518	$43,021	**0.872**
Allen University	SC	$30,497	$34,290	**0.889**
Provo College	UT	$39,645	$41,733	**0.950**
Provo College-Idaho Falls Campus	ID	$39,645	$41,733	**0.950**
Laurus College	CA	$30,896	$32,416	**0.953**

4. Anomaly Detection

A single metric misses institutions that perform poorly across multiple dimensions simultaneously. An **Isolation Forest** model was trained on four features — value score, median debt, completion rate, and admission rate — to flag multi-dimensional outliers.

Parameter	Value
Algorithm	Isolation Forest (scikit-learn)
Features	value_score, median_debt, completion_rate, admission_rate
Contamination	5% expected outlier rate
Scaling	StandardScaler applied before fitting
Flagged	69 institutions after data quality filtering
Filters	completion_rate > 0 and student_size >= 100

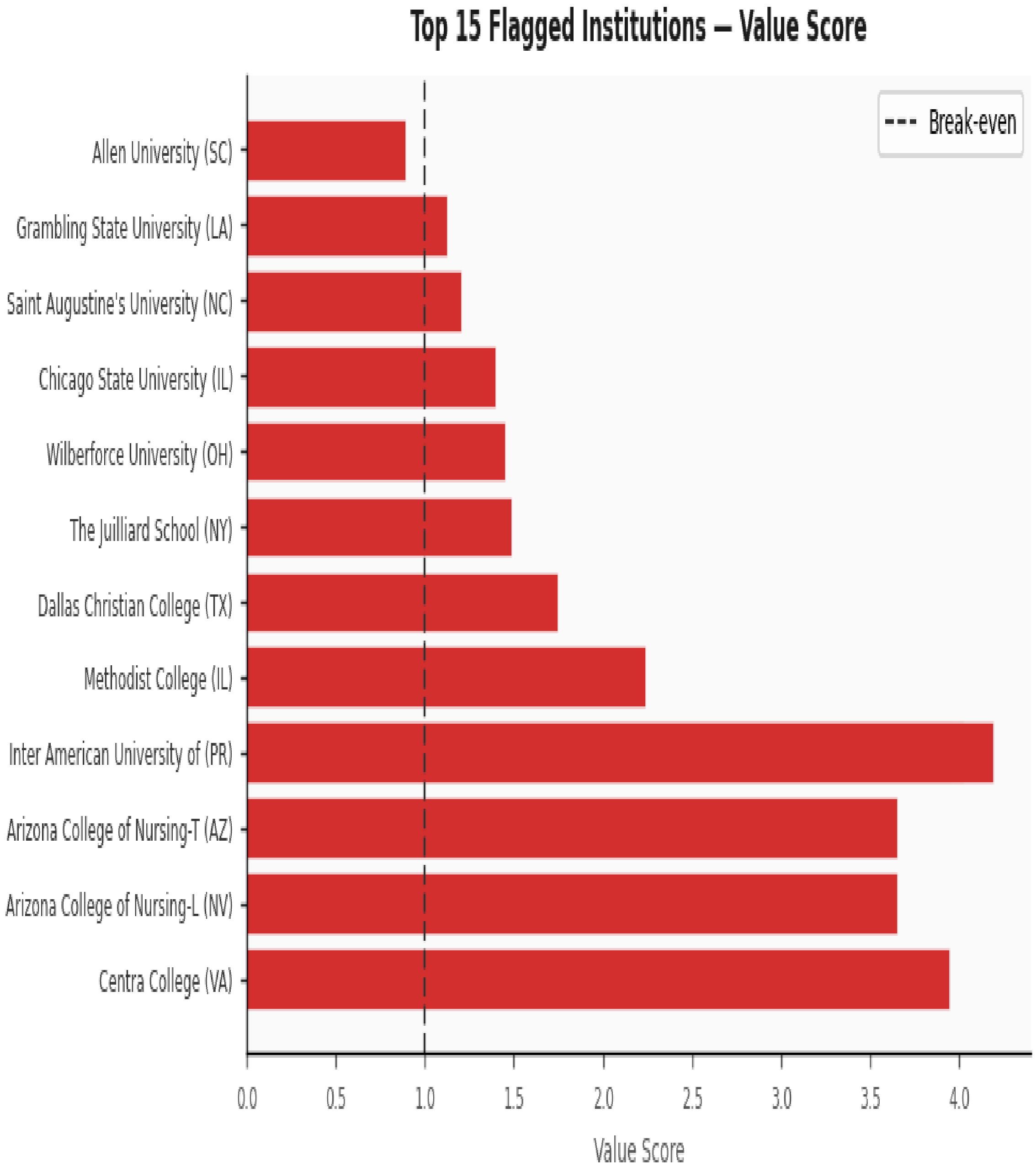


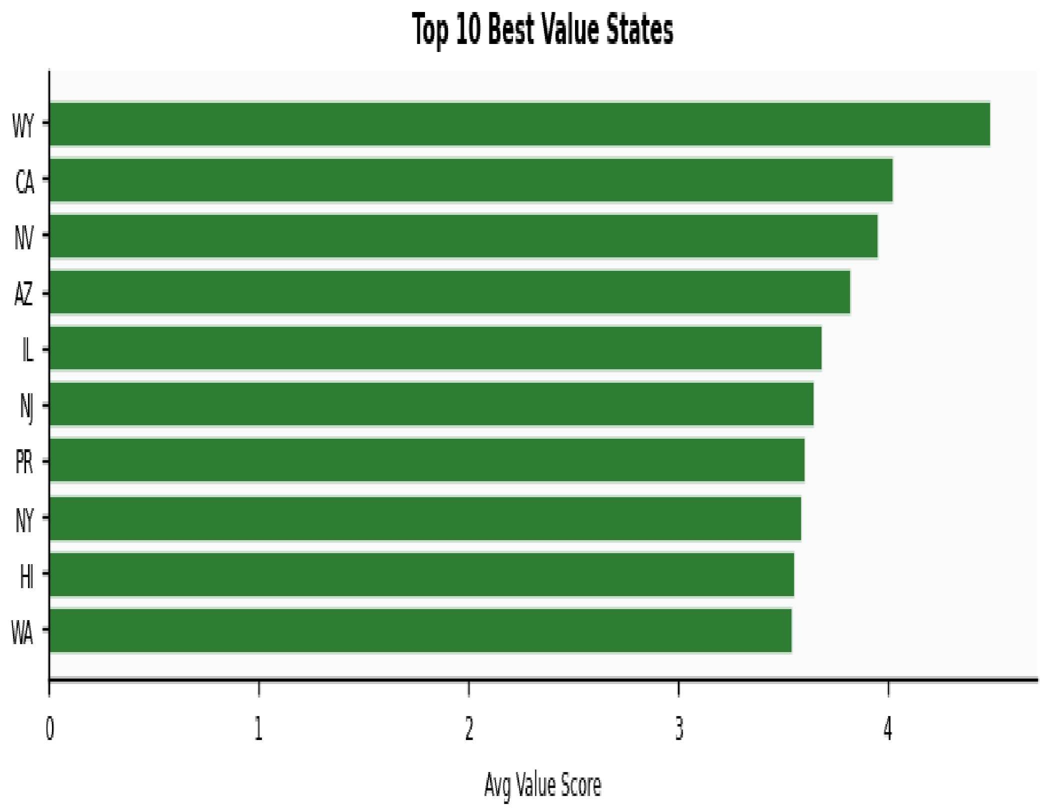
Figure 3 — Flagged as outliers across all four dimensions simultaneously.

Data Quality Nuance

Several flagged institutions have acceptable value scores but zero completion rates — reflecting newly launched programs with no graduates yet, not poor outcomes. Addressed via quality filtering.

5. State-Level Analysis

Value scores vary significantly by state, reflecting differences in tuition, local labor markets, cost of living, and institution mix.



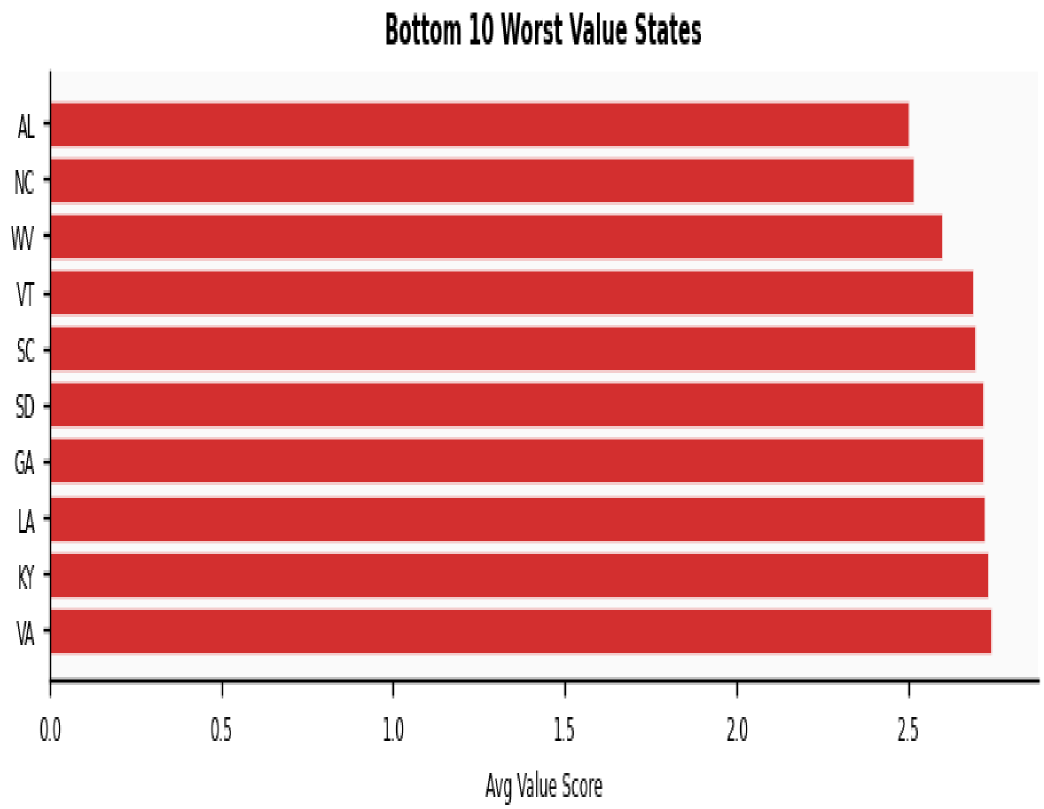


Figure 4 — Best and worst value states. States with fewer than 3 institutions excluded.

• **New Jersey ranks 6th nationally** (avg 3.64). Top 2 individual value schools in the country are both NJ institutions.

• **Wyoming leads nationally** — low state tuition plus strong energy-sector wages.

• **Puerto Rico ranks lowest** — structural economic conditions, not institutional quality. A geographic bias in the metric.

• **School size is irrelevant** — r = 0.09 correlation between enrollment and value score.

6. How This Helps Students

The interactive Streamlit dashboard translates 4,500 rows of federal data into four practical tools.

Tool	What It Does	Who It Helps
Value Rankings	Sort 4,500 schools by earnings/debt, filter by state	Students comparing schools locally
Flagged List	69 schools flagged as multi-dimensional outliers	Anyone doing due diligence
Plain English Queries	Ask questions in plain English — Claude API converts to SQL	Parents, counselors, non-technical users
State Comparison	Compare all 50 states with school-level drill-down	Students considering out-of-state options

Limitations

- **10-year earnings median** — early career looks different in some fields.
- **Institution-level debt** — not program-level; medical and liberal arts programs share the same figure.
- **Correlation not causation** — low score reflects outcomes, not instruction quality.
- **Geographic wage bias** — lower-wage states produce lower scores regardless of school quality.

7. Summary of Findings

#	Finding	Implication
1	Mean value score 3.22 across 4,500 institutions	Most colleges are financially sound investments
2	46 schools below break-even (score < 1.0)	Graduates earn less than borrowed — concentrated risk
3	69 institutions flagged as multi-dimensional outliers	Poor across debt, earnings AND completion simultaneously
4	NJ ranks 6th nationally; top 2 schools both in NJ	Strong local value without leaving state
5	School size: r = 0.09 correlation with value	Brand and enrollment are poor proxies for ROI
6	Flagged schools cluster at $5K-$25K debt	Moderate cost can still mean poor outcomes
7	Puerto Rico has highest flagged concentration	Geographic wage gaps create structural bias

This analysis demonstrates that the college debt crisis, while real, is not uniformly distributed. The federal data exists to identify exactly which institutions are leaving students worse off — and this tool makes that identification accessible to anyone.